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A Randomized Trial of a Low-Fat Dietary Intervention in Women at High Risk for Breast Cancer

Michael S. Simon, Lance K. Heilbrun, Allison Boomer, Cynthia Kresge, Jan Depper, Paik N. Kim, Frederick Valeriote, and Silvana Martino

Abstract: A randomized intervention trial of dietary fat reduction to 15% of total calories was initiated in 1987 for women at high risk for breast cancer to determine the feasibility of recruiting and maintaining them on a low-fat diet. The study has enrolled 194 women between the ages of 18 and 67 years who met at least one of three eligibility criteria: 1) a first-degree relative with breast cancer, 2) a P2 or DY Wolfe mammographic pattern, and 3) a prior breast biopsy demonstrating epithelial hyperplasia with or without atypia. Eligible women must also have had diets that contained $\geq 30\%$ of calories from fat at entry. Women were randomized to a nonintervention usual diet vs. a 15% low-fat diet. Recruitment was sought through physicians, personal mailings, breast cancer patients, and the news media. Two study sites participated: a large urban hospital affiliated with a university medical center and a community oncology private practice. The results from both institutions were similar and demonstrated that a low-fat dietary plan could be effectively conducted in private as well as academic settings with recruitment tailored to the community where the trial is being conducted. Reduction in dietary fat intake was maximal during the first three months of the dietary intervention and remained stable throughout 12 months of follow-up. Reductions in total calories, weight loss, and percent body fat were minimal. The nonintervention group experienced no major change in their diet. We conclude that it is feasible to recruit women who are at high risk for breast cancer into a dietary intervention trial, and with sufficient dietary counseling and motivation on the part of participants, reduction in dietary fat intake can be achieved and maintained. More in-depth analyses of these data will be presented in subsequent reports.

Introduction

Despite recent improvements in the early detection and treatment of breast cancer, breast cancer mortality rates in

the United States have remained essentially unchanged among Caucasian women and have increased by approximately 1% per year among African-American women since 1973 (1,2). Whereas it is hoped that routine screening mammography and the use of adjuvant therapy may ultimately have an impact on survival, the results are modest and have prompted us to redirect our efforts toward breast cancer prevention.

In 1987 we started a breast cancer prevention program at a large academic medical center and at a community hospital. The purpose of the program was to study the effect of a low-fat diet (LFD) intervention in a group of women at high risk for breast cancer. Our goals were to determine whether women at high risk for breast cancer could be recruited to participate in a research project that would involve considerable time and effort and to determine the feasibility of reducing the fat intake of women in the intervention group to 15% of their total calories. In this report, we describe the methods and results of recruitment, dietary counseling strategies, and changes in a number of parameters related to dietary intake achieved during a 12-month period.

Methods

Recruitment of Subjects

Our plan was to recruit to a breast cancer prevention program 200 women who were ≥ 18 years of age and who were at high risk for developing breast cancer. Eligibility criteria included 1) a family history of breast cancer in a mother, sister, or daughter, 2) a P2 or DY Wolfe classification pattern on a mammogram (3), and 3) epithelial hyperplasia with or without atypia on a previous breast biopsy. Women who were pregnant or lactating or who were planning a pregnancy in the next two years were excluded. Women were also excluded if they were an insulin-dependent or unstable diabetic or if they were on any hormones,

anticonvulsants, phenothiazine derivatives, antihypertensive agents, or cholesterol-lowering agents. Approval for the study was obtained from the Institutional Review Board of both institutions.

We used a variety of strategies to recruit participants to the diet study. Women were recruited from the Breast Cancer Prevention Program at Harper Hospital, which is affiliated with the Karmanos Cancer Institute and Wayne State University (WSU; Detroit, MI). Women were also recruited from the Hospital Corporation of America Wesley Hospital (HCA-WH; Wichita, KS). Recruitment efforts at WSU included in-house advertising to physicians and cancer patients, large mailings to University and Medical Center employees and to the Women's Economic Club of Detroit, and major media presentations (newspaper articles and television news features) at area health fairs. The HCA-WH participated in a week-long series of local television programs on breast cancer, during which the diet study was discussed. In addition, there were meetings with community physician groups and articles in various hospital publications for the Kansas site. Participant registration began in April 1987 and ended in August 1989 at WSU and began in February 1988 and ended in May 1989 at HCA-WH.

Eligibility Screening and Randomization

At both research sites, initial eligibility screening took place over the telephone. If the potential candidate was determined to be eligible, a physician visit was scheduled for a medical history and physical examination and review of prior mammograms and/or breast biopsies. Patient informed consent was obtained at the time of the initial visit. Once clinical eligibility was established, a three-day food record was obtained to determine dietary eligibility. Subjects were given instructions on how to complete the three-day food records themselves. Women were eligible for the study if their diet contained $\geq 30\%$ of calories from fat. An analysis of food records for eligibility was performed using an in-house microcomputer system (Nutritionist III, N-Squared Computing, Silverton, OR), which provided rapid determination of dietary eligibility. After eligibility was established, the WSU participants were scheduled for a baseline blood sample and needle aspirate of each breast, after which women were randomized to the LFD or nonintervention (NI) arm of the study. Recruitment at HCA-WH differed in that the initial visit occurred with the oncology nurse and dietitian followed by an appointment with a family physician for a physical examination. No breast needle aspirates were done on HCA-WH women.

Dietary randomization took place at WSU. A two-arm randomization between NI or LFD was performed using a block size of two. Before randomization, women were stratified by age at entry into three groups: <35 , 35–50, and ≥ 51 years. The NI women were asked to maintain their usual food intake, whereas the LFD women were taught to reduce their dietary fat intake to 15% of total calories. Subjects

were asked to make at least a one-year commitment to the study. After this proved feasible, we later asked new subjects for a two-year commitment and those completing their first year to continue for a second year.

Clinical and Laboratory Assessments

Participants underwent a standard history and physical examination and baseline mammogram with repeat physical examinations at six-month intervals and repeat annual mammography for two years. Height and weight, percent body fat, and serology (lipids, prolactin, and estrogen) were measured at baseline, at three-month intervals for Year 1, and at six-month intervals for Year 2. Percent body fat was estimated using the technique of body composition analysis based on bioelectrical impedance (4). Waist and hip measurements were obtained at baseline and then at six-month intervals for two years. Breast needle aspiration was performed at baseline and at yearly intervals (only on WSU women). Results of laboratory assays measured on subsamples of women from this study have been reported elsewhere (5–7). We used an interviewer-administered questionnaire to obtain information on subject employment, stress factors, smoking history, exercise habits, weight, diet, gynecological and reproductive history, prior breast disease, oral contraceptive or other hormone use, previous radiation exposure, and family history of cancer. Questionnaire results were directly entered into a computer during the subject interview.

Dietary Counseling and Data Collection

Participants in the LFD group were counseled by registered dietitians to follow a low-fat eating plan, developed by the Nutrition Coordinating Center at the University of Minnesota and by the American Health Foundation (8). The eating plan is a step-by-step individualized approach to reducing total fat intake using a combination of education, goal setting, evaluation, feedback, and participant self-monitoring. Participants met with a dietitian biweekly for three months of intensive one-on-one counseling followed by monthly visits in an individual or a group format. Participants were switched to monthly meetings if they completed a minimum of eight weeks of food diaries, met their fat gram goal five of seven days (from their food diaries), had a fat intake between 15% and 20%, and met individual dietary goals 75% of the time. After the three-month intensive intervention, women with fat intakes $>20\%$ of calories from fat were contacted to arrange for additional individual counseling sessions.

Special topics covered in the monthly group meetings included eating away from home, family involvement, social support, cooking with dried beans and peas, food cues, packing lunches, low-fat desserts, fast and easy cooking, and what to do when you "slip." Discussions also focused on interpersonal support and sharing information on new restaurants serving low-fat foods and new products or recipes.

Women in the NI group followed their own diets and received no counseling regarding fat intake. They were given a copy of the US Department of Agriculture Dietary Guidelines, which suggest moderate fat intake; however, this group did not receive any education on how to meet these recommendations. Both groups were monitored for the nutritional adequacy of their diets in terms of 1989 Recommended Dietary Allowance values. If their diet, based on their three-day dietary records (collected at baseline and at 3-mo intervals), was <67% of the Recommended Dietary Allowance for a specific vitamin or mineral, then they met with the dietitian and received written information about specific nutrient deficiencies.

Three-day food records were used to assess dietary intake at baseline and three-month intervals, alternating Sunday/Monday/Tuesday with Thursday/Friday/Saturday, to give a more representative sampling of days. If a food record was not received within two weeks of the assigned date, the subject was contacted. Dietary data that could not be collected within 30 days of the assigned date were considered missing. Twenty-four-hour food recalls were collected at three-month intervals by appointment or by telephone. Additional dietary data were collected annually for both groups using the Harvard Food Frequency Questionnaire (9). Study dietitians maintained frequent telephone contacts between sites to maintain the continuity of the intervention. In addition, the principal investigator and project coordinator from WSU visited the HCA-WH site.

Dietary Analysis

The dietary records and recalls were sent to the Nutrition Coordinating Center at the University of Minnesota. Nutrient intake data were received on magnetic tape within three months. Individual progress reports of nutrient intake were prepared quarterly from the three-day food records by the study programmer/analyst and became part of the participant's file to be evaluated by the dietitian. The reports included information on fat intake (percentage of total calories from fat and average fat intake in g/day), average calories per day, and vitamin and mineral intake. This information was used to monitor compliance with the LFD. The annual food frequency questionnaires were analyzed by the Channing Laboratory at the Harvard Medical School. Dietary and laboratory data were maintained on ORACLE (version 6.0, ORACLE, Belmont, CA). Statistical analysis was performed using SAS (version 6.08, SAS Institute, Cary, NC).

Results

Description of Study Participants

Women were referred to the program from a variety of sources. Television accounted for 30% of referrals in Detroit and 34% in Wichita. In Detroit, a local television station broadcast a feature story about the Breast Cancer Prevention

Program, and of 406 telephone calls and letters received in response, 31 women were randomized into the study. The next largest source of referral in Detroit was the newspaper (22%). A feature article about the program published in the largest Detroit area newspaper generated 203 calls and letters, from which 24 women were randomized into the study. For women at the Wichita site, a physician (25%) or family member (19%) was the next largest source of referral. Eight percent of women from both sites combined learned about the program from a brochure, from direct mail, or from friends.

A total of 265 women were evaluated for eligibility (185 WSU and 80 HCA-WH). Of these, 205 (77%) were eligible and 194 (73%) were randomized and enrolled. Participation rates were similar for both sites. The most common reason for ineligibility was failure to return the three-day food record (38%) followed by a diet that was already <30% in fat content (33%). Reasons for nonparticipation among 11 eligible women included inability to make the time commitment (6), attempting pregnancy (2), relocation (1), refusal of the mammogram (1), and failure to return calls to the program (1).

Table 1 shows the age distribution of women participating in the study by institution. Both institutions were most successful in recruiting women between 35 and 50 years of age. Table 2 shows the demographic characteristics of 188 women participating in both sites combined (6 women failed to complete the health status questionnaire). The distribution of demographic characteristics was similar for women at both sites. The participants were 89% Caucasian, 9% African-American, and 2% Hispanic. The majority of women (64%) were married, 82% were college graduates or had some college, and 80% were employed outside the home.

In terms of risk factor eligibility, family history of breast cancer (in a first-degree relative) accounted for 81% of all participants. Fifty-two percent of the women ≥ 35 years of age had a P2 or DY mammographic pattern, whereas only 8% had epithelial hyperplasia with or without atypia on prior breast biopsies (not all women who were initially screened for participation had a prior breast biopsy). Thirty-nine percent of the participants had two of three eligibility criteria risk factors and 1% had all three risk factors.

Table 1. Age Distribution of Participants in the Breast Cancer Prevention Program^a

Age, ^b yrs	WSU (n = 137)		HCA-WH (n = 57)	
	n	%	n	%
<35	20	14	16	28
35–50	90	66	25	44
≥ 51	27	20	16	28

a: A total of 194 women were followed; WSU, women recruited from Breast Cancer Prevention Program at Harper Hospital (affiliated with Karmanos Cancer Institute and Wayne State University, Detroit, MI); HCA-WH, women recruited from Hospital Corporation of America Wesley Hospital (Wichita, KS).

b: Age at randomization.

Table 2. Demographic Characteristics of Participants in the Breast Cancer Prevention Program^a

Demographic Characteristics	n	%
Race		
Caucasian	168	89
African-American	17	9
Hispanic	3	2
Marital status		
Single	34	18
Married	121	64
Divorced	4	2
Separated	24	13
Widowed	5	3
Education		
Grade 7-9	1	1
Grade 10-12	2	1
High school graduate	31	16
Some college	64	34
College graduate	40	21
Some graduate school	16	9
Graduate degree	34	18
Employment		
Working full-time	111	59
Working part-time	40	21
Homemaker	27	14
Retired	5	3
Unemployed	5	3

a: A total of 188 women participated; health status questionnaire was not completed for 6 women.

Dietary Data

Dietary data are based on assessments of three-day food records. The mean intake of selected nutrients stratified by diet group site is described in Table 3. There were no apparent differences in nutrient intake by study site, and therefore data for both sites are presented together. For descriptive statistical purposes, dietary data are presented for all women that contributed information on nutrient intake for any of the five time points in the study.

At baseline, the NI group had a somewhat lower total caloric intake and total fat intake than the LFD group. During the 3-month intensive intervention, participants in the LFD group lowered their mean percent caloric intake from fat from $36.0 \pm 0.7\%$ (SE) to $19.0 \pm 0.7\%$, and this change was maintained at 12 months into the study. Similar reductions can be seen with regard to total fat intake in grams.

Both diet groups had a reduced mean caloric intake at 12 months compared with baseline. Dietary fiber intake in the LFD group increased at 3 months, and this increase was maintained at 12 months, whereas for the NI group the dietary fiber intake decreased slightly.

A small decrease in body weight occurred for the LFD and the NI group during the 12 months of follow-up (Table 4). Women in the LFD group lost an average of 3 pounds during the first year of the study, and women in the NI group lost an average of 5 pounds. Percent body fat (as measured by the electrical impedance method) decreased slightly for both groups during the study year.

Table 3. Mean Intake of Selected Dietary Variables for the Breast Cancer Prevention Program Women by Diet Arm^a

	Low-Fat Diet		Nonintervention Diet	
	n	Mean $\pm$ SE	n	Mean $\pm$ SE
Calories				
Baseline	93	$1,876 \pm 66.6$	91	$1,676 \pm 49.6$
3 mos	77	$1,559 \pm 40.1$	85	$1,608 \pm 46.2$
6 mos	68	$1,563 \pm 36.6$	82	$1,537 \pm 51.6$
9 mos	66	$1,650 \pm 50.6$	72	$1,515 \pm 62.9$
12 mos	65	$1,570 \pm 47.0$	68	$1,594 \pm 63.6$
%Calories from fat				
Baseline	93	36.0 ± 0.7	91	35.9 ± 0.7
3 mos	77	19.0 ± 0.7	85	34.7 ± 0.8
6 mos	68	17.1 ± 0.5	82	34.4 ± 0.9
9 mos	66	17.3 ± 0.6	72	34.1 ± 0.9
12 mos	65	18.0 ± 0.7	68	33.8 ± 0.9
Total fat, g				
Baseline	93	78.2 ± 3.8	91	67.9 ± 2.6
3 mos	77	32.5 ± 1.2	85	63.7 ± 2.5
6 mos	68	29.8 ± 1.2	82	60.3 ± 2.8
9 mos	66	31.5 ± 1.4	72	59.2 ± 3.3
12 mos	65	31.4 ± 1.4	68	61.4 ± 3.4
Dietary fiber, g				
Baseline	93	16.5 ± 0.6	91	16.2 ± 0.7
3 mos	77	19.1 ± 1.0	85	16.1 ± 0.7
6 mos	68	20.5 ± 0.8	82	15.0 ± 0.7
9 mos	66	20.7 ± 0.9	72	15.4 ± 0.8
12 mos	65	20.2 ± 0.9	68	15.4 ± 0.8

a: Data are based on 3-day food records from all women available at each time point of study; n, no. of women.

Description of Dropouts

Of the 47 dropouts from both locations, 26 were in the LFD group and 21 were in the NI group (Table 5). Among the LFD group, the three reasons most often cited for dropping out were illness or death of family member, inability to maintain the time commitment, and unwillingness to fol-

Table 4. Mean Weight and Percent Body Fat of Breast Cancer Prevention Program Women by Diet Arm and by Time on Study^a

	Low-Fat Diet		Nonintervention Diet	
	n	Mean $\pm$ SE	n	Mean $\pm$ SE
Body weight, pounds				
Baseline	98	149.4 ± 3.5	96	155.6 ± 3.2
3 mos	80	148.0 ± 3.8	89	153.2 ± 3.3
6 mos	74	146.2 ± 3.9	85	154.6 ± 3.5
9 mos	67	140.7 ± 2.8	76	150.9 ± 3.1
12 mos	67	146.1 ± 4.4	76	150.3 ± 2.7
%Body fat				
Baseline	89	31.9 ± 0.6	87	33.3 ± 0.6
3 mos	76	31.8 ± 0.6	87	32.8 ± 0.6
6 mos	73	30.9 ± 0.6	82	33.0 ± 0.6
9 mos	67	30.5 ± 0.7	75	32.9 ± 0.6
12 mos	67	30.7 ± 0.7	76	32.8 ± 0.6

a: Data are from all women available at each time point of study; n, no. of women.

Table 5. Distribution of Women That Dropped Out of the Breast Cancer Prevention Program by Diet Arm and Reason for Dropout

Reason	Low-Fat Diet (<i>n</i> = 26)		Nonintervention Diet (<i>n</i> = 21)	
	<i>n</i>	%	<i>n</i>	%
Illness or death of family member	5	19	3	14
Inability to maintain time commitment	5	19	6	28
Personal problems	3	11.5	2	10
Unwillingness to follow eating plan	5	19	3	14
Loss of interest in program	1	4	2	10
Relocation	2	18	12	10
Change of eligibility status	2	8	2	10
Other	3	11.5	1	5

low the eating plan. Among the NI women, the time commitment was the single largest reason for dropping out.

Standard survival data analysis of the time on study showed very comparable results by institution (Figure 1; $p = 0.635$ by log-rank test). At 12 months the cumulative proportions of women still on study at WSU and HCA-WH were 76% and 69%, respectively. When the institutions were combined, the 12-month study retention rates were 81% on NI and 69% on LFD (75% overall), and time on study differed significantly by diet group (Figure 2; $p = 0.009$ by log-rank test).

Discussion

Cancer prevention studies present many new challenges to medical oncologists whose background is rooted in treatment-related clinical trials. Perhaps the greatest challenge resides in the fact that prevention trials may involve healthy individuals believed to be at risk for cancer, rather than patients with cancer. To initiate a dietary intervention for women at high risk for developing breast cancer, we assumed that there would be many women in our community

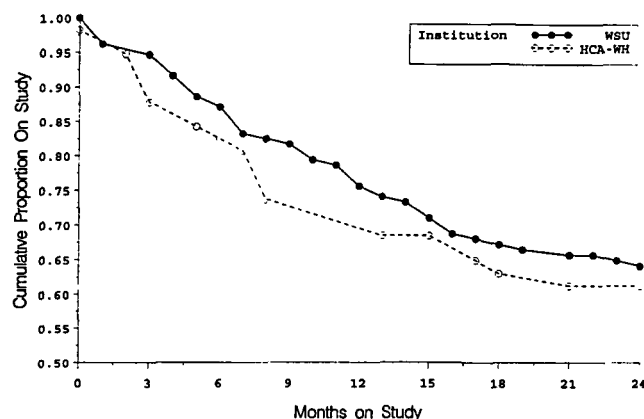


Figure 1. Breast Cancer Prevention Program cumulative proportion of subjects on study by institution.

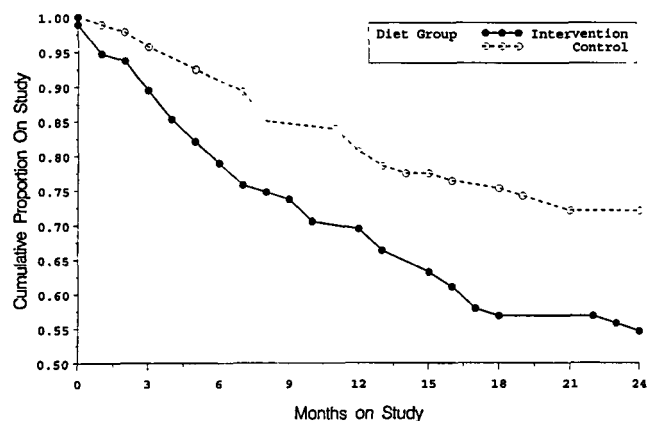


Figure 2. Breast Cancer Prevention Program cumulative proportion of subjects on study by diet group. Intervention, low-fat diet (LFD) group; control, nonintervention (NI) group.

who fit our categories of high risk. We also assumed that women who were perceived to be at a higher-than-average risk of developing breast cancer were seeking medical attention from family physicians, gynecologists, and breast cancer-screening centers and that these physicians would welcome a research project where they could refer these women.

During the time that our program was active, a variety of strategies were used for recruitment. Advertising to our own breast cancer patients and to physicians within our own institution did not yield the necessary accrual; therefore, a number of recruitment efforts were made in the community. Some of these included large mailings and major media efforts. Participants were sought through two major metropolitan Detroit newspaper articles, several smaller community newspapers, four television news spots, and several radio spots. Word of mouth from program participants or their family and friends was also a source of referral. In addition, we established a relationship with a private oncologist in a community-based hospital in Wichita, KS, who through his own efforts contributed subjects.

We found that television and newspaper articles were our best source of subjects. Small local newspapers and radio were less effective, and even fewer participants were referred by physicians. A similar approach with good results from media efforts has been reported by Halper and co-workers (10). Additionally, they reported good results by placing a recruitment message in a New York Telephone Company mailing. Perhaps our biggest surprise has been the observation that, despite our having a large population of women with breast cancer who are cared for in our own institution and to whom our program was advertised, few of their family members enrolled in the intervention trial.

Another factor affecting recruitment was seasonal variation. The lowest number of new participants was recruited in the summer, whereas autumn was the most successful time. We also learned that because this was a research project, some of the subjects assumed that they would be paid, and some expressed anger that doctor visits and mammo-

grams were billed as routine clinical services. Whereas women were informed during the consenting process that they would receive no free medical care or monetary compensation, it was our impression that many women perceived themselves as volunteers and therefore felt that they were not responsible for clinical services received.

Demographic data on our patients revealed that our subjects were predominantly well educated, employed, and self-referred, which is similar to subject characteristics observed in other trials of dietary and chemopreventive agents (11). We were unsuccessful in attracting minority participants, which limits the generalizability of the results and suggests that perhaps specifically tailored recruitment strategies are necessary for that goal to be accomplished (12). Both institutions were most successful attracting women between 35 and 50 years of age.

Family history of breast cancer was the most frequent risk factor identified in our cohort, likely reflecting the fact that most of our participants learned about our program through the media and were self-referred. Of the three high-risk criteria, family history is the only one that women could easily identify about themselves, whereas mammographic and histological criteria would require input from a physician.

The success of our dietary counseling was equally apparent in both research sites. The major reduction in percent dietary fat took place within the first 3 months of the study, and both dietary groups maintained a stable pattern throughout the first 12 months of follow-up. It was not our goal to have women reduce their fat intake to <15% of total calories, and women who were observed to drop below this level were counseled to bring their fat intake level back up to 15%. Whereas some of the study participants were interested in weight reduction, this was not encouraged. The reduction in fat intake observed was accompanied by a minimal reduction in total calories, weight, and percent body fat. These changes took place predominantly during the first three months of the study and remained stable thereafter.

Of particular concern to us was whether the NI group would alter its eating habits. Our dietary counseling was designed to provide limited education to this group. However, dietary information is easily available to the general public, a factor over which we would have no control. We also considered that simply being a part of a dietary research program might influence eating habits. A minimal reduction in dietary fat and total calories did occur in the NI group and resulted in a small change in body weight and percent body fat. From this observation, it appears that unless an educational program is provided, simply providing general literature on good nutrition is insufficient to reduce dietary fat intake.

Short-term feasibility of reducing fat intake in postmenopausal women with breast cancer has been reported. In 18 patients, Boyar and co-workers (13) were able to demonstrate a reduction of fat intake from $34 \pm 7\%$ (SD) to $22 \pm 6\%$ over a six-month period. Similarly, in 17 patients, Buzzard and associates (8) reported a reduction in mean daily fat intake from $38.4 \pm 4.3\%$ to $22.8 \pm 7.8\%$ at three

months. More recently, Chlebowski and others (14) reported a reduction in dietary fat for women participating in the Women's Intervention Nutrition Study. The fat intake for 143 women in the dietary intervention arm (15% of caloric intake from fat) was reduced from $34.0 \pm 6.1\%$ to $20.3 \pm 2.4\%$ at 3 months and maintained throughout 24 months of observation.

A larger study with longer follow-up has been reported by Boyd and colleagues (15). Women with mammographic dysplasia were randomly allocated to a control group or a 15% low-fat diet group. Both groups consumed diets at entry containing 37% of calories from fat. The control group reduced their intake to 36% by 4 months and remained at that level through 12 months. The low-fat group reduced fat intake to 23% by four months and remained at that level for one year. A recent follow-up of this cohort showed that, at almost eight years after intervention, women in the low-fat group had a significantly lower fat intake than the control women (16).

The Women's Health Trial has reported data at one and two years of follow-up. Initially designed as a multiinstitutional trial to evaluate the role of a low-fat diet on reduction of breast cancer incidence in women at high risk, it accrued 303 volunteers randomized to a usual vs. a low-fat diet. One year data published by Gorbach and co-workers (17) demonstrated that the intervention group was able to reduce its fat intake to 22% by 6 months and maintain that level for 12 months. Subsequent publications by Henderson and associates (18) and Insull and others (19) have reported that these levels are maintained to 24 months and for up to one year after completion of the trial (20). Clearly, women from our cohort, the Women's Health Trial participants, women with mammographic dysplasia, and women with breast cancer are selected highly motivated groups of women with a concern that they may develop breast cancer or suffer a recurrence and may not be representative of the general population. Renewed interest in the issue of dietary fat and breast cancer has been recently expressed by the National Cancer Institute advisors (21). A new Women's Health Initiative has been started. The trial's hypothesis is that dietary fat reduction will reduce not only the incidence of breast cancer but also colorectal cancer, coronary heart disease, and total mortality.

In conclusion, our data indicate that 1) recruitment of women at increased risk for breast cancer to a dietary intervention trial can be accomplished, 2) recruitment efforts need to be tailored to the community where the study is being conducted, 3) the public media can be an aid in recruitment, and 4) given a certain level of interest, educational techniques can result in a marked reduction in dietary fat intake for a prolonged period of time. What remains unresolved is whether these new patterns once learned become permanent and whether changes in dietary fat intake modulate intermediate markers of breast cancer risk. Ultimately, if a decrease in breast cancer incidence is observed in women on an LFD intervention, new LFD public educa-

tion strategies need to be developed to reduce cancer incidence in the general population.

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